



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

17-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	A	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C	A,B,D	A,C	A,C	B,C	A,B,C	A,B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.50	7.00 to 7.06	3.10 to 3.20	1.40 to 1.43	7.00	3.75 to 3.85		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	A	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,D	A,B	A,B,C,D	A,B,C,D	A,B	B,C	A,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	6.00	5.00	1044.40	0.16 or 0.17	1.50		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	B	D	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C	A,B,C,D	A,B,D	A,C,D	A,C,D	A,C,D	A,C	B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.00	0.14	0.52 to 0.53	1.00	2.71 to 2.72	4.24		

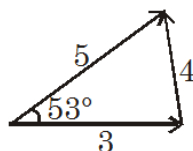
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PART-1 : PHYSICS

SECTION-I (i)

2. Ans (B)

$$i_3 = i_1 + i_2$$



3. Ans (A)

$$a = A \sin(\omega t_0)$$

$$b = A \sin(2\omega t_0)$$

$$c = A \sin(3\omega t_0)$$

$$a + c = 2A \sin 2\omega t_0 \cos \omega t_0$$

$$\frac{a + c}{2b} = \cos \omega t_0$$

$$\Rightarrow f = \frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a + c}{2b} \right)$$

4. Ans (A)

$$\vec{B} = B_0 \hat{i} [\cos(kz - \omega t)] + B_1 \hat{j} \cos(kz + \omega t)$$

$$B_0 = 3 \times 10^{-5} \text{ T } B_1 = 2 \times 10^{-6} \text{ T}$$

Electric field associated with it is,

$$\vec{E} = -B_0 c \hat{j} \cos(kz - \omega t) - B_1 c \hat{i} \cos(kz + \omega t)$$

Here, c is speed of light in vacuum.

At $z = 0$,

$$F_{\text{rms}} = \sqrt{\frac{(QB_0 c)^2 + (QB_1 c)^2}{2}}$$

Here c is speed of light in vacuum.

$$= \sqrt{\frac{(10^{-4} \times 3 \times 10^{-5} \times 3 \times 10^8)^2 + (10^{-4} \times 2 \times 10^{-6} \times 3 \times 10^8)^2}{2}}$$

$$= \sqrt{\frac{0.81 + 0.0036}{2}} = 0.6 \text{ N}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,B,C)**

Brewster's law

6. **Ans (A,B,D)**

$$Q = Q_0 \sin \omega t$$

$$F = \frac{Q^2}{2\epsilon_0 A} = \frac{Q_0^2 \sin^2 \omega t}{2\epsilon_0 A}$$

$$F_{av} = \frac{Q_0^2}{4\epsilon_0 A} = \frac{400 \times 10^{-12}}{4 \times 40 \times 10^{-6} \times 10^4}$$

$$= \frac{1}{40} \text{N} = 25 \text{mN}$$

$$U_B = \frac{1}{2} Li^2 = \frac{1}{2} L \times Q_0^2 \omega^2 \cos^2 \omega t$$

$$U_{av} = \frac{1}{2} L \times Q_0^2 \times \frac{1}{LC} \times \frac{1}{2}$$

$$= \frac{Q_0^2}{4C} = \frac{400 \times 10}{4 \times 40 \times 10^{-6}} = 2.5 \mu\text{J}$$

7. **Ans (A,C)**

$$(a) 100 = \omega = \sqrt{\omega_0^2 - \gamma^2}$$

$$\omega_0 = 125 \text{ rad/s}$$

$$\frac{R}{2L} = \gamma = \sqrt{125^2 - 100^2} = \sqrt{25 \times 225} \text{ rad/s}$$

$$= 75 \text{ rad/s}$$

$$R = 150 \Omega$$

$$(b) i = \frac{V}{R} = \frac{200}{150} \text{A} = 1.33 \text{A}$$

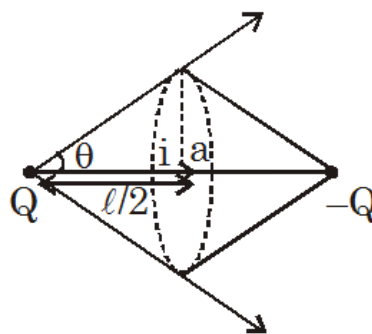
$$(c) \omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow C = \frac{1}{125^2} = 64 \mu\text{F}$$

$$(d) P = \frac{V^2}{Z^2} R = \frac{200^2}{\left(\omega L - \frac{1}{\omega C}\right)^2 + R^2} \times R$$

$$= \frac{4 \times 10^4 \times 150}{\left(100 - \frac{10^4}{100 \times 64}\right)^2 + (150)^2} = \frac{4 \times 10^4 \times 150}{100^2 + 150^2}$$

$$= \frac{400 \times 150}{325} = \frac{60000}{325} \text{W} = \frac{2400}{13} \text{W}$$

8. **Ans (A,C)**



$$\phi_E = \frac{Q}{2\epsilon_0} (1 - \cos \theta) \times 2$$

$$\epsilon_0 \frac{d\phi_E}{dt} = i_d ; i_d = \frac{dQ}{dt} (1 - \cos \theta)$$

Since Q is decreasing $= i(1 - \cos \theta)$

i_d is to left.

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 (i + i_d) = \mu_0 [i - i(1 - \cos \theta)]$$

$$B \times 2\pi a = \mu_0 i \cos \theta$$

$$= \mu_0 i \times \frac{\ell/2}{\sqrt{a^2 + \frac{\ell^2}{4}}}; B = \frac{\mu_0 i \ell}{4\pi a \sqrt{a^2 + \frac{\ell^2}{4}}}$$

9. **Ans (B,C)**

$$F = -QE_0 \hat{i} \sin \omega t = m \frac{dv}{dt}$$

$$v = -\frac{QE_0}{m\omega} \cos \omega t \Big|_0^t = \frac{QE_0}{m\omega_0} (1 - \cos \omega t) \hat{j}$$

$$\vec{B} = \frac{E_0}{c} \hat{k} \sin(kx - \omega t)$$

at $x = 0$

$$\vec{B} = -\frac{E_0}{c} \hat{k} \sin(\omega t)$$

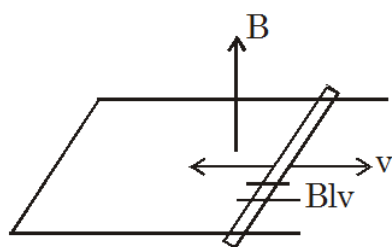
$$\vec{F} = Q\vec{v} \times \vec{B} = -\frac{Q^2 E_0^2}{m\omega_0 c} \hat{i} (\sin \omega t - \sin \omega t \cos \omega t)$$

$$F_{av} = \frac{Q^2 E_0^2}{m\omega_0 c} \frac{\cos 2\omega t}{\frac{2\omega}{T}} \Big|_0^T = 0$$

10. **Ans (A,B,C)**

$$Z = \frac{n\lambda}{d} \sqrt{D^2 + x^2}$$

11. Ans (A,B,C)



$$i = \frac{Blv}{R}$$

$$F = -Bil - kx = ma$$

$$a + \frac{B^2 \ell^2 v}{mR} + \frac{kx}{m} = 0$$

$$\frac{d^2x}{dt^2} + \frac{B^2 \ell^2}{mR} \frac{dx}{dt} + \frac{k}{m}x = 0$$

$$\omega_0 = \sqrt{\frac{k}{m}}; \gamma = \frac{B^2 \ell^2}{2mR}$$

$$\omega = \sqrt{\frac{k}{m} - \frac{B^4 \ell^4}{4m^2 R^2}}$$

$$= \sqrt{\frac{50}{0.4} - \frac{0.4^4 \times 0.5^4}{4 \times 0.4^2 \times 0.01^2}}$$

$$= 10 \text{ rad/s} = 1 \text{ me}^{-\frac{B^2 \ell^2}{2mR} \times \frac{2\pi}{10}}$$

$$= 1 \text{ e}^{-\frac{0.5^2 \times 0.4 \times \pi}{0.4 \times 0.01 \times 10}} = \frac{1}{e^\pi}$$

12. Ans (A,C,D)

When a compression pulse strikes a fixed surface, it reflects as compression pulse.

PART-1 : PHYSICS

SECTION-II

1. Ans (2.50)



$$U + K = E$$

$$mgy + \frac{1}{2}mv^2 = E$$

$$\frac{mgx^2}{4a} + \frac{1}{2}mv^2 = E$$

$$\frac{dE}{dt} = 0 \Rightarrow \frac{mgx}{2a}v + mv \frac{dv}{dt}$$

$$\frac{dv}{dt} = -\frac{gx}{2a}; \omega = \sqrt{\frac{g}{2a}}$$

$$= \sqrt{\frac{100}{1.6}} = \frac{10}{4} = 2.5 \text{ rad/s}$$

2. Ans (7.00 to 7.06)

$$i = \frac{V_0}{\sqrt{x_C^2 + R^2}} = \frac{V_0}{\sqrt{x_C^2 + \left(\frac{V}{i}\right)^2}}$$

$$x_C^2 = \frac{V_0^2}{i^2} - \frac{V^2}{i^2} \Rightarrow x_C = \frac{V_0^2 - V^2}{i^2}$$

$$\frac{1}{\omega C} = \sqrt{\frac{V_0^2 - V^2}{i^2}}$$

$$\Rightarrow C = \frac{i}{\omega \sqrt{V_0^2 - V^2}}$$

$$= \frac{0.28}{2\pi \times 50 \sqrt{127^2 - 3.5^2}} = 7.02 \mu\text{F}$$

3. Ans (3.10 to 3.20)

$$I = \frac{p_0^2}{2\rho V} = \frac{p_0^2 v}{2 \times B}$$

$$10^{-8} = \frac{p_0^2 \times 320}{2 \times 1.6 \times 10^5}$$

$$p_0^2 = \frac{3.2 \times 10^{-4}}{320} = 10^{-5}$$

$$p_0 = \sqrt{10} \times 10^{-3} \text{ N/m}^2$$

$$= 3.16 \times 10^{-3} \text{ N/m}^2$$

4. Ans (1.40 to 1.43)

$$I = \frac{P}{A} = \frac{1}{2} \epsilon_0 E_0^2 c$$

$$\therefore E_0 = \sqrt{\frac{2P}{\epsilon_0 c A}} = \sqrt{\frac{2 \times (27 \times 10^{-3})}{(9 \times 10^{-12}) \times (3 \times 10^8) \times (10 \times 10^{-6})}}$$

$$\Rightarrow E_0 = 1.4 \text{ kV/m}$$

5. Ans (7.00)

Let v_1 and v_2 be the velocities of sources, then apparent frequencies are

$$f_1 = \left(\frac{c + v_1}{c - v_1}\right) f_0 \text{ and } f_2 = \left(\frac{c + v_2}{c - v_2}\right) f_0$$

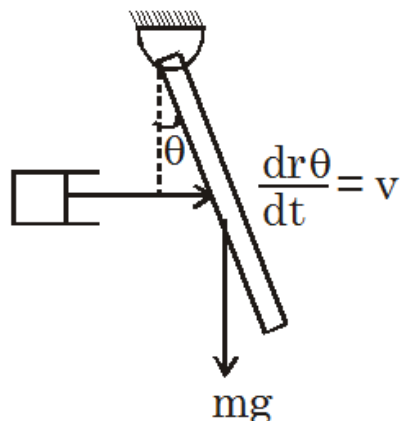
Then difference between frequencies is,

$$\Delta f = \frac{2\Delta v}{C} f_0$$

$$\therefore c \gg v_1 \text{ \& } c \gg v_2$$

$$\therefore \Delta v = 7 \text{ kmph}$$

6. Ans (3.75 to 3.85)



$$\tau_{\text{net}} = -mg \times \frac{\ell}{2} \sin \theta - C\theta \times r + \ell F$$

$$I \frac{d^2\theta}{dt^2} + Cr^2 \frac{d\theta}{dt} + mg \frac{\ell}{2} \theta = \ell F \sin(\omega t)$$

$$\frac{d^2\theta}{dt^2} + \frac{3Cr^2}{m\ell^2} \frac{d\theta}{dt} + \frac{3mg\ell}{2m\ell^2} \theta = \frac{3F\ell}{m\ell^2} \sin \omega t$$

$$A = \frac{3F}{m\ell\omega} \times \frac{1}{\sqrt{\left(\omega - \frac{3g}{2\ell}\right)^2 + \left(\frac{3Cr^2}{m\ell^2}\right)^2}}$$

$$= \frac{3 \times 10}{10 \times 2 \times 20} \times \frac{1}{\sqrt{\left(20 - \frac{3 \times 10}{2 \times 20}\right)^2 + \left(3 \times \frac{4 \times 1^2}{10 \times 2}\right)^2}}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (B)

$$t = 20 \text{ min}, \theta = 40^\circ$$

$$t = 50 \text{ min}, \theta = 10^\circ$$

$$\Rightarrow t_{1/2} = 15 \text{ min.}$$

$$\text{at } t = 60 \text{ min.} \Rightarrow 4 \text{ half life}$$

$$\Rightarrow \text{Volume of } \text{H}_2\text{O}_2 \text{ consumed}$$

$$= 40 + 20 + 10 + 5 = 75 \text{ ml.}$$

3. Ans (A)

Virial equation of state is

$$Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \dots (1)$$

An equivalent form of the virial expansion is an infinite series in power of the pressure.

$$\frac{PV_m}{RT} = 1 + B'P + C'P^2 + \dots (2)$$

$$\text{From (1), } \frac{PV_m}{RT} = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

$$\text{or, } P = \frac{RT}{V_m} \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right) (3)$$

From (1) and (2)

$$1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots = 1 + B'P + C'P^2 + \dots (4)$$

From (3) and (4)

$$1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots = 1 + B' \frac{RT}{V_m} + C' \frac{R^2 T^2}{V_m^2} + \dots$$

$$\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)^2 + \dots$$

$$\text{Or, } 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots = 1 +$$

$$\frac{B'RT}{V_m} + \frac{BB'RT}{V_m^2} + \frac{C'(RT)^2}{V_m^2} + \dots$$

$$\text{Or, } 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots = 1 +$$

$$\frac{B'RT}{V_m} + \frac{RT}{V_m^2} (BB' + C'RT) + \dots (5)$$

Both sides of equation (5) are power series in

$\frac{1}{V_m}$. Since the two power series must be equal,

the coefficient of each power of $\frac{1}{V_m}$ must be

the same on both sides.

$$\therefore B = B'RT \Rightarrow B = \frac{B}{RT} \dots (6)$$

$$\text{and } C = BB'RT + C'(RT)^2$$

$$\text{Or, } C = B \times \frac{B}{RT} \times RT + C'(RT)^2$$

$$\text{Or, } C = B^2 + C'(RT)^2$$

$$\text{Or, } C = \frac{C - B^2}{(RT)^2} \dots (7)$$

PART-2 : CHEMISTRY

SECTION-I (ii)

6. **Ans (A,B)**

Trans alkene + Br₂ in CCl₄ $\xrightarrow{\text{Anti addition}}$
Meso product (A & B)

8. **Ans (A,B,C,D)**

Vf=1 Vf=1

(Na₂CO₃ + NaHCO₃) → 10 ml solution

↓ ↓

x mmole y mmole

At HPh end point,

eq. of HCl = eq. of Na₂CO₃

$$\frac{1}{5} \times 2.5 \times 10^{-3} \times 1$$

$$= x \times 10^{-3} \times 1$$

$$\Rightarrow x = 0.5 \text{ mmole} = 0.5 \times 10^6 \text{ mg} = 53 \text{ mg} = 0.053 \text{ g}$$

At Mo end point

eq. of HCl = 2 × eq. of Na₂CO₃ + eq. of NaHCO₃

$$\frac{1}{5} \times 7.5 \times 10^{-3} \times 1 = (x \times 10^{-3} \times 1) \times 2 + y \times 10^{-3} \times 1$$

$$\Rightarrow 1.5 = 1 + y \Rightarrow y = 0.5 \text{ mmole}$$

$$= 0.5 \times 84 = 42 \text{ mg} = 0.042 \text{ mg}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.00)**

Geometrical centers = 3 and compound is symmetric then number of geometrical isomers

$$= 2^{n-1} + 2^{\frac{n-1}{2}} = 6$$

3. **Ans (5.00)**

P = KX

$$1 = K \times \frac{1 \times 40}{\frac{300R \times 1000}{\frac{100}{18}}}$$

$$\Rightarrow 1 = K \times \frac{40}{300R} \times \frac{18}{100} \times \frac{1}{1000}$$

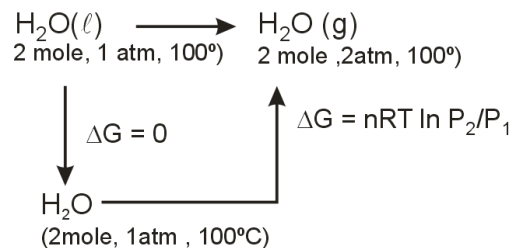
$$\Rightarrow K = \frac{300R \times 100 \times 1000}{40 \times 18}$$

P = KX

$$\text{or, } 4 = \frac{100 \times 300R}{18 \times 40} \times \frac{x}{48 \times \frac{1600}{18}}$$

$$\text{or, } x = \frac{4 \times 40 \times 48 \times 16}{300R \times 1000} = 5\text{g}$$

4. **Ans (1044.40)**



$$\Delta G = 2 \times 2 \times 373 \ln \frac{2}{1}$$

$$= 4 \times 373 \times 0.7 = 1044.4 \text{ cal.}$$

PART-3 : MATHEMATICS

SECTION-I (i)

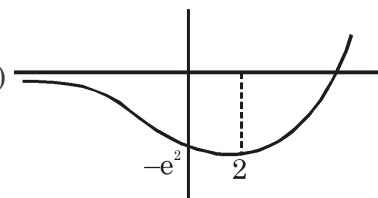
1. **Ans (C)**

$$e^x(x-3) = k$$

$$\text{Let } f(x) = e^x(x-3)$$

$$f'(x) = e^x(x-2)$$

$$k \in (-e^2, 0)$$



2. **Ans (B)**

$$[x]^2 = x + 2\{x\} = [x] + 3\{x\}$$

$$\{x\} = \frac{[x]^2 - [x]}{3} \Rightarrow (0 \leq \{x\} < 1)$$

$$\Rightarrow 0 \leq \frac{[x]^2 - [x]}{3} < 1$$

$$\Rightarrow [x] \in \left(\frac{1-\sqrt{3}}{2}, 0 \right] \cup \left[1, \frac{1+\sqrt{3}}{2} \right)$$

$$\Rightarrow [x] = -1, 0, 1, 2$$

$$\text{or } \{x\} = \frac{2}{3}, 0, 0, \frac{2}{3}$$

$$\Rightarrow x = -\frac{1}{3}, 0, 1, \frac{8}{3}$$

3. **Ans (D)**

$$\lim_{x \rightarrow 0} \frac{(1+3x+2x^2)^{1/x} - e^3}{x}$$

$$- \lim_{x \rightarrow 0} \frac{(1+3x-2x^2)^{1/x} - e^3}{x} = L_1 - L_2$$

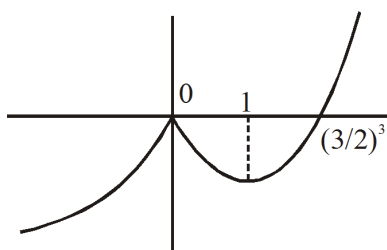
(say)

$$L_1 = \lim_{x \rightarrow 0} \frac{e^{\frac{\ln(1+3x+2x^2)}{x}} - e^3}{x} = \frac{-5}{2} e^3$$

$$L_2 = \frac{-13}{2} e^3$$

4. Ans (D)

Graph of $y = 2x - 3x^{2/3}$



PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C)

$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$(x, y) = (a \cos^3 \theta, a \sin^3 \theta)$$

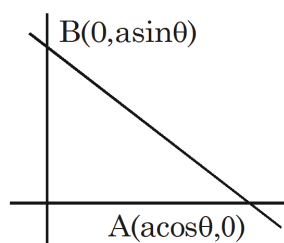
$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\frac{\sin \theta}{\cos \theta}$$

$$(y - a \sin^3 \theta) = -\frac{\sin \theta}{\cos \theta} (x - a \cos^3 \theta)$$

$$x \sin \theta + y \cos \theta = a \sin \theta \cos \theta$$

$$\frac{x}{\cos \theta} + \frac{y}{\sin \theta} = a$$

$$\frac{x}{\left(\frac{x_1}{a}\right)^{1/3}} + \frac{y}{\left(\frac{y_1}{a}\right)^{1/3}} = a \Rightarrow \frac{x}{x_1^{1/3}} + \frac{y}{y_1^{1/3}} = a^{2/3}$$



$$AB = \sqrt{a^2 \cos^2 \theta + a^2 \sin^2 \theta} = a$$

$$2h = a \cos \theta, 2k = a \sin \theta$$

$$4x^2 + 4y^2 = a^2$$

equation of normal

$$x^{2/3} + y^{2/3} = a^{2/3} \Rightarrow \frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} y' = 0$$

$$y' = -\frac{y^{1/3}}{x^{1/3}}$$

$$(y - y_1) = \frac{x_1^{1/3}}{y_1^{1/3}} (x - x_1)$$

$$x_1^{1/3} x - y y_1^{1/3} = x_1^{4/3} - y_1^{4/3}$$

6. Ans (A,B,C,D)

$$2|f(x) - f(y)| \leq |x - y|$$

$$\Rightarrow \frac{|f(x) - f(y)|}{|x - y|} \leq \frac{1}{2}$$

$$\lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \frac{1}{2}$$

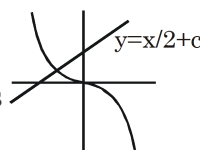
$$|f'(y)| \leq \frac{1}{2}$$

$$\Rightarrow f'(x) \in \left[-\frac{1}{2}, \frac{1}{2} \right] \text{ and } f'(x) \geq \frac{1}{2}$$

from both we can say that $f'(x) = \frac{1}{2}$

$$\Rightarrow f(x) = \frac{x}{2} + C$$

(A) $y = \frac{x}{2} + C$ and $y = x$ is



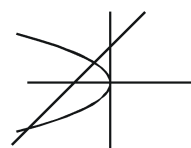
one

(B) $y = \frac{x}{2} + C$ and $y = -x^3$ is one

(C) $y = \frac{|x|}{2}$ and $y = \frac{x}{2} + C$

can be one, infinite or zero.

(D) $y^2 = -x$



may be more than one

7. Ans (A,B,D)

$$f(x) = (x-1) \tan^{-1} x - \frac{1}{2} \ln(1+x^2)$$

$$f'(x) = \tan^{-1} x - \frac{1}{1+x^2}$$

$$f'(x) > 0 \quad \forall x \geq 1$$

$\Rightarrow f(x)$ is increasing function

$$f''(x) > 0$$

$\Rightarrow f'(x)$ is increasing function $\forall x \geq 1$

$$\lim_{x \rightarrow \infty} f'(x) \rightarrow \frac{\pi}{2} \Rightarrow f'(x) < \frac{\pi}{2} \quad \forall x \geq 1$$

Using LMVT in $[x, x+4]$, $x \geq 1$

$$\frac{f(x+4) - f(x)}{x+4-x} = f'(c) \text{ for } c \in (x, x+4)$$

$$\therefore f(x+4) - f(x) < 2\pi \quad \forall x \in [1, \infty)$$

8. Ans (A,C,D)

$$f(x) = \begin{cases} 1 & x \in (2n\pi, 2n\pi + \frac{\pi}{2}) \quad n \in \mathbb{I}^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x = (4n+1)\frac{\pi}{2} \quad n \in \mathbb{I}^+ \cup \{0\} \\ \frac{5\log_e x}{2} & x \in (2n\pi + \frac{\pi}{2}, 2n\pi + \frac{3\pi}{2}) \quad n \in \mathbb{I}^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x = 2n\pi + \frac{3\pi}{2} \quad n \in \mathbb{I}^+ \cup \{0\} \\ 1 & x \in (2n\pi + \frac{3\pi}{2}, 2n\pi + 2\pi) \quad n \in \mathbb{I}^+ \cup \{0\} \end{cases}$$

$f(x)$ is discontinuous at all positive odd integral multiple of $\frac{\pi}{2}$

$f(x)$ is continuous at integral multiple of π

9. Ans (A,C,D)

For A, D, assume $g(x) = x^n f(x)$

For C; apply LMVT in $[0, 4]$ and $[4, 5]$

10. Ans (A,C,D)

$$f(0) = 0, f(1) = 2, f(2) = 4$$

Using L.M.V.T on $[0, 1]$

$\exists$ a point $x = \alpha \in (0, 1)$, such that

$$f'(\alpha) = \frac{f(1) - f(0)}{1 - 0} = 2$$

Similarly apply L.M.V.T. on $y = f(x)$ in $[1, 2]$

$\exists$ a point $x = \beta \in (1, 2)$, such that

$$f'(\beta) = \frac{f(2) - f(1)}{2 - 1} = 2$$

Now, Applying Rolle's theorem on $y = f'(x)$

$$x \in [\alpha, \beta]$$

$\therefore f'(\alpha) = f'(\beta)$ and other conditions are satisfied

$\exists$ a point $x = \gamma \in (\alpha, \beta)$ $f''(\gamma) = 0$ for some

$$\gamma \in (0, 2)$$

11. Ans (A,C)

$$P_n = \lim_{x \rightarrow 0^+} \log \left(\frac{\tan x}{x} \cdot \frac{\tan 2x}{x} \cdot \frac{\tan 3x}{x} \cdots \frac{\tan nx}{x} \right)$$

$$= \log n!$$

$$P_{256} - P_{255} = \log 256 = 8 \log 2!$$

$$\therefore k = 8 \text{ and } n = 2$$

12. Ans (B,C,D)

Since both the curves touches each other

$$\Rightarrow m_1 = m_2$$

$$4(x-a) = 2e^{2x}$$

$$4(x-a) = 2.2(x-a)^2$$

$$\Rightarrow (x-a)(x-a-1) = 0$$

$$\Rightarrow x = a \text{ or } x = a + 1$$

If $x = a$ then (I^{st}) curve $\Rightarrow y = 0$

& 2^{nd} curve $\Rightarrow y = e^{2a}$ but $e^{2a} \neq 0 \forall a \in \mathbb{R}$

$$\therefore x \neq a$$

$$\Rightarrow x = a + 1$$

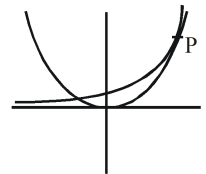
$$\therefore 1^{st} \text{ curve } \Rightarrow y = 2 \quad \therefore (x, y) = (a + 1, 2)$$

(x, y) lies on 2^{nd} curve also, $\therefore 2 = e^{2a+2}$

$$\Rightarrow 2a + 2 = \ln 2$$

$$a = \frac{1}{2}(\ln 2 - 2)$$

$$\Rightarrow -1 < a < 0$$



PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$f(x) = |(2^x - 1) \cdot (3^x - 3)x \cdot (x-1) \cdot (x-2)|$$

$f(x)$ is not differentiable at $x = 2$

2. Ans (0.14)

$$S_n = \sum_{k=0}^n \tan^{-1} \frac{49}{49k^2 + 63k + 57} = \sum_{k=0}^n \tan^{-1} \frac{1}{k^2 + \frac{63}{49}k + \frac{57}{49}}$$

$$= \sum_{k=0}^n \tan^{-1} \frac{1}{1 + \left(k + \frac{8}{7}\right) \left(k + \frac{1}{7}\right)}$$

$$= \sum_{k=0}^n \left(\tan^{-1} \left(k + \frac{8}{7}\right) - \tan^{-1} \left(k + \frac{1}{7}\right) \right)$$

$$\Rightarrow S_n = \tan^{-1} \left(n + \frac{8}{7}\right) - \tan^{-1} \frac{1}{7}$$

$$S_{\infty} = \cot^{-1} \frac{1}{7}$$

3. Ans (0.52 to 0.53)

$$\begin{aligned}\frac{dy}{dx} &= \left((\sin^{-1}x)^2 - \pi (\cos^{-1}x) \right) \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \left((\sin^{-1}x)^2 - \pi \left(\frac{\pi}{2} - \sin^{-1}x \right) \right) \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \left((\sin^{-1}x)^2 + \pi \sin^{-1}x - \frac{\pi^2}{2} \right) \frac{1}{\sqrt{1-x^2}} \\ &= \left(\left(\sin^{-1}x + \frac{\pi}{2} \right)^2 - \frac{3\pi^2}{4} \right) \frac{1}{\sqrt{1-x^2}} \\ \frac{dy}{dx} &= 0 \Rightarrow \sin^{-1}x + \frac{\pi}{2} = \pm \frac{\sqrt{3}}{2}\pi \\ \sin^{-1}x &= \sqrt{3}\frac{\pi}{2} - \frac{\pi}{2} \text{ only} \\ \Rightarrow \frac{\pi}{2} - \cos^{-1}\alpha &= \sqrt{3}\frac{\pi}{2} - \frac{\pi}{2} \\ \Rightarrow \pi - \cos^{-1}\alpha &= \sqrt{3}\frac{\pi}{2} \\ \Rightarrow \frac{\cos^{-1}(-\alpha)}{\pi} &= \frac{\sqrt{3}}{2}\end{aligned}$$

4. Ans (1.00)

$$f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} \quad (f(0) = 0)$$

$$f'(x) = 1 + x + x^2 + x^3 + x^4 \quad (f'(0) = 1)$$

$$f''(x) = 1 + 2x + 3x^2 + 4x^3 \quad (f''(0) = 1)$$

$$\text{Now } f(g(x)) = x$$

$$\text{Diff. w.r.to } x$$

$$f'(g(x))g'(x) = 1 \quad \dots(i)$$

$$\text{again Diff. w.r.to } x$$

$$f''(g(x))(g'(x))^2 + g''(x)f'(g(x)) = 0$$

$$x = 0$$

$$f''(g(0))(g'(0))^2 + g''(0)f'(g(0)) = 0 \quad \dots(ii)$$

$$\text{as } f(0) = 0 \Rightarrow g(0) = 0$$

$$\text{and from (i) } g'(0) = \frac{1}{f'(0)}$$

$$\text{now from (ii)}$$

$$f''(0)\left(\frac{1}{f'(0)}\right)^2 + g''(0)f'(0) = 0$$

$$1.1^2 + g''(0).1 = 0$$

$$g''(0) = -1$$

$$|g''(0)| = 1$$

5. Ans (2.71 to 2.72)

$$a_1 = 1, a_n = n(1 + a_{n-1})$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{a_1}\right) \left(1 + \frac{1}{a_2}\right) \dots \left(1 + \frac{1}{a_n}\right)$$

$$\lim_{n \rightarrow \infty} \frac{(1 + a_1)(1 + a_2) \dots (1 + a_n)}{a_1 a_2 a_3 \dots a_n}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{\left(\frac{a_2}{2}\right) \left(\frac{a_3}{3}\right) \dots \left(\frac{a_{n+1}}{n+1}\right)}{a_1 a_2 a_3 \dots a_n} = \frac{a_{n+1}}{(n+1)!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{(n+1)(1 + a_n)}{(n+1)!} = \frac{1}{n!} + \frac{a_n}{n!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{n(1 + a_{n-1})}{n!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$$

Continuing in same way;

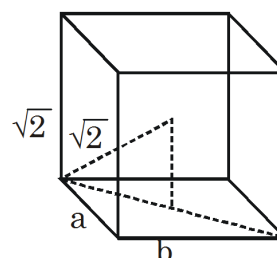
$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{1!} + \frac{a_1}{1!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} \dots + \frac{1}{n!} \quad (\text{In reverse order})$$

As $n \rightarrow \infty$, series on RHS converges to e

6. Ans (4.24)

Let the other edges be 'a' & 'b'



$$\left(\frac{\sqrt{a^2 + b^2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 = 2$$

$$\frac{a^2 + b^2}{4} = \frac{3}{2} \Rightarrow a^2 + b^2 = 6$$

Now, volume of cuboid = $ab\sqrt{2}$

$$\therefore \frac{a^2 + b^2}{2} \geq \sqrt{a^2 b^2}$$

$$\frac{a^2 + b^2}{2} \geq ab \Rightarrow ab \leq 3$$

$$\therefore V_{\max} = 3\sqrt{2}$$